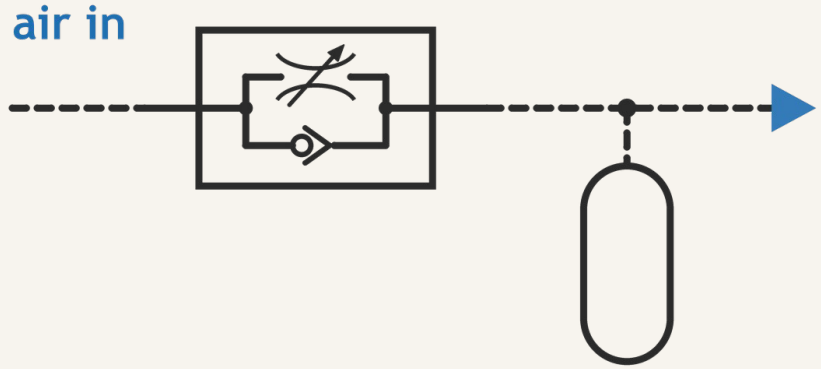
Time Delays

The glue needs time to set. How does a press know when to let go?



National 5 Engineering Science

2019 N5 Engineering Science: scan the code and try these.

Q10(c)

Page 17. Answer in your jotter.

Scan for the paper

Marking scheme

2025 N5 Engineering Science: scan the code and try these.

Q2(a)

Page 3. Answer in your jotter.

Scan for the paper

Marking scheme

PART 1

How a Time Delay Works

To describe how a uni-directional restrictor and a reservoir make a time delay.

By the end of this part… (p.16)



1. I can **describe** how a uni-directional restrictor and a reservoir make a time delay.
2. I can **draw** a uni-directional restrictor and reservoir the right way round.
3. I can **describe** how to make a time delay longer or shorter.
4. I can **describe** the operation of a circuit that has a time delay.

The Reservoir (p.16)

A **reservoir** is an empty tank. Air must fill it before the pressure in it builds up.

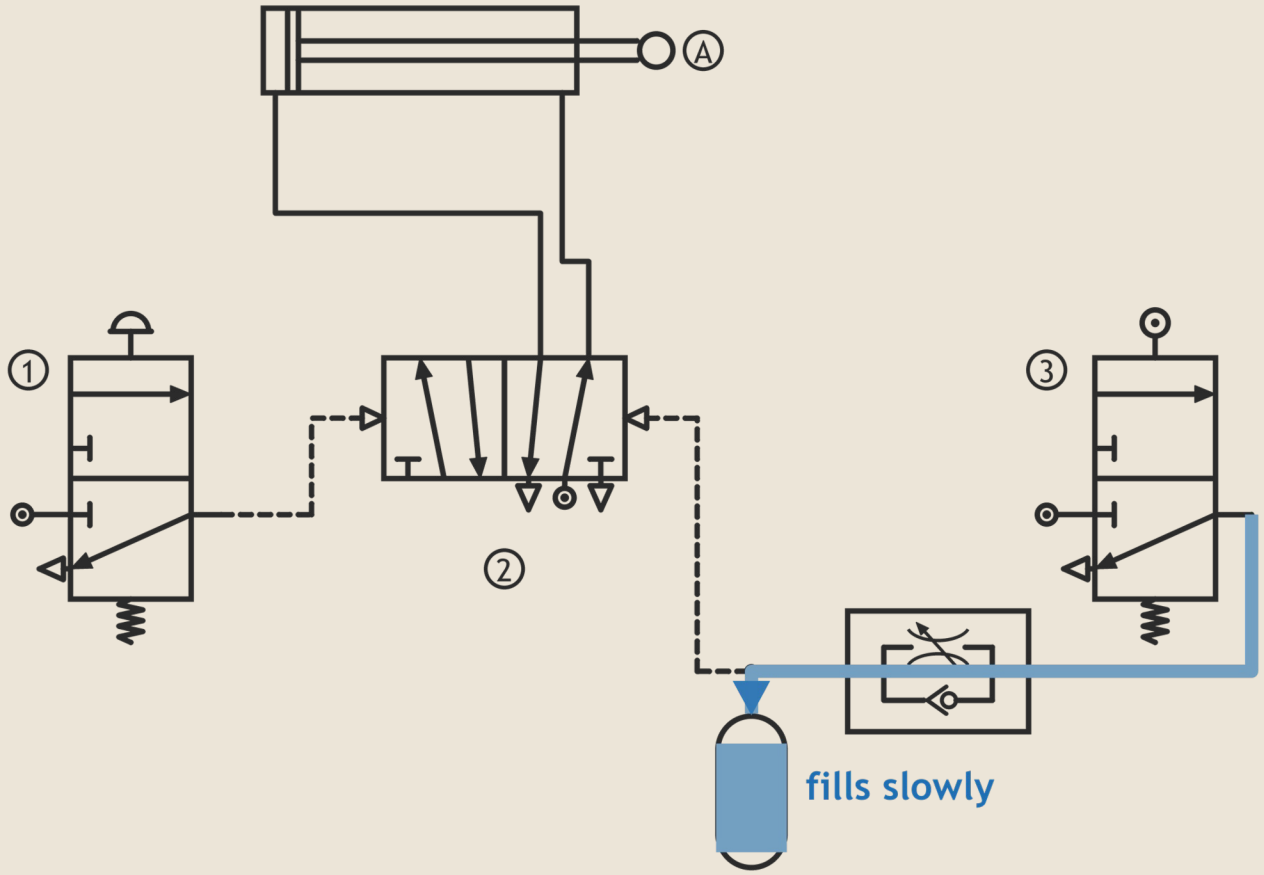
Its symbol is a **long round-ended** shape.



How a Time Delay Works (p.16)

A uni-directional restrictor slows the air going **into** the reservoir, so the reservoir fills slowly.

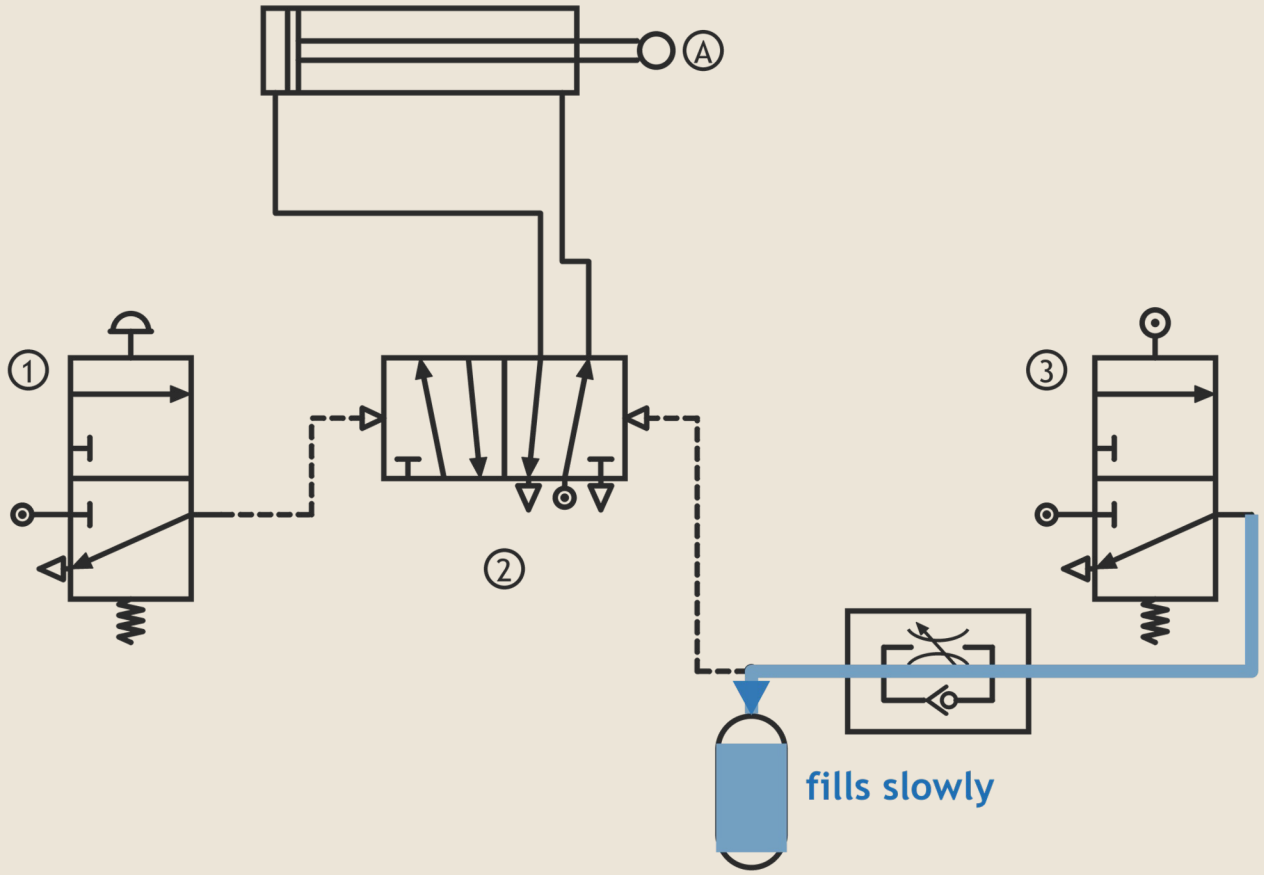
When the pressure is high enough, the pilot air **changes the state** of the valve.



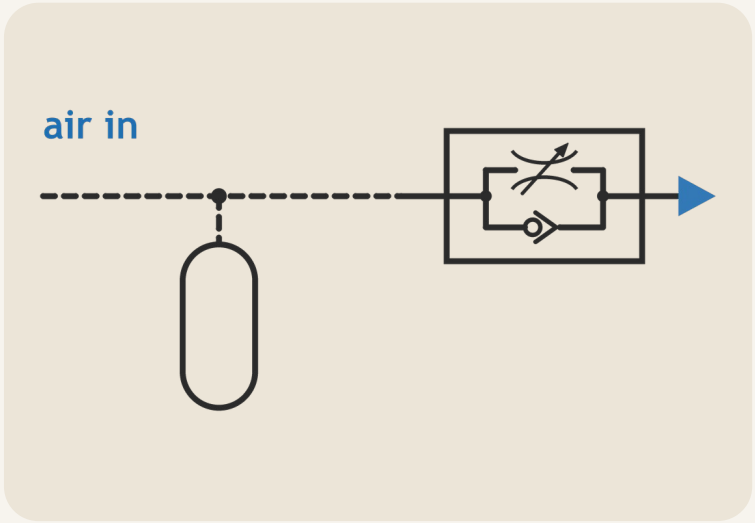
The Time Delay (p.16)

The time the reservoir takes to fill is the **time delay**.

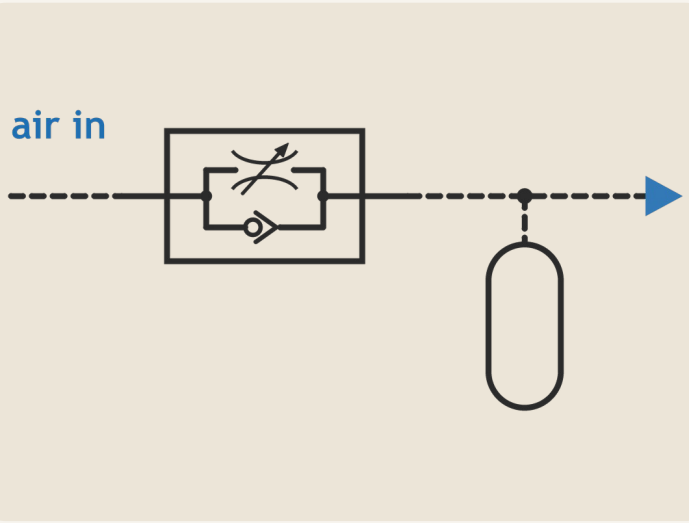
The arrow on the restrictor means the time delay is **adjustable**.



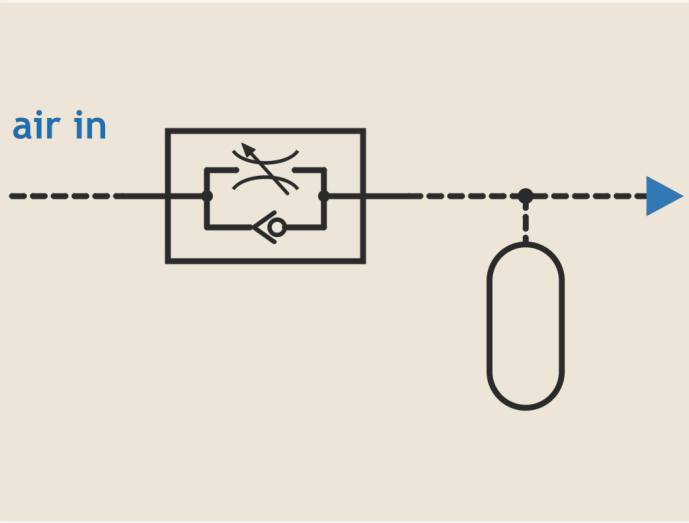
Show Me: Which One Makes a Delay?



No: the reservoir is **before** the restrictor



Yes: the restrictor slows the air **into** the reservoir

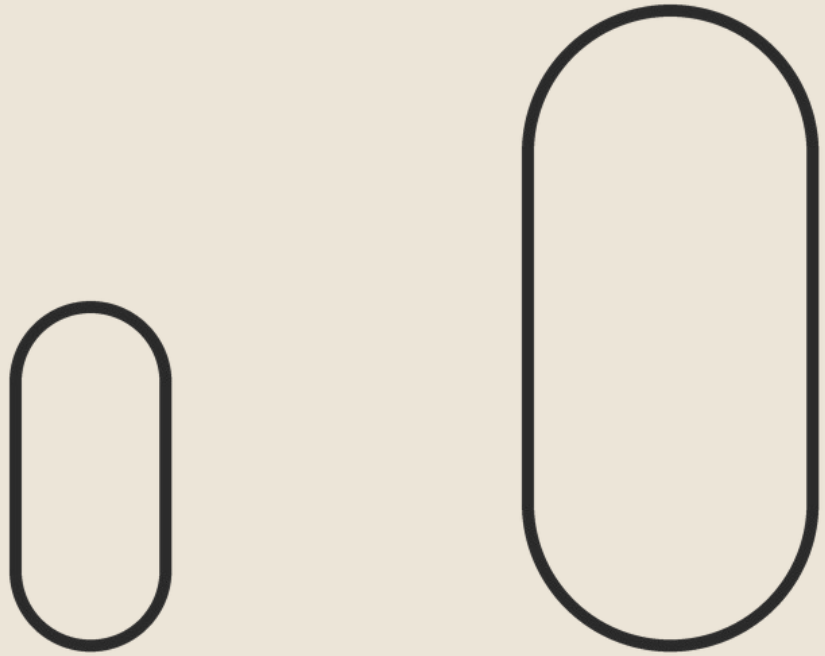


No: the restrictor is the **wrong way round**

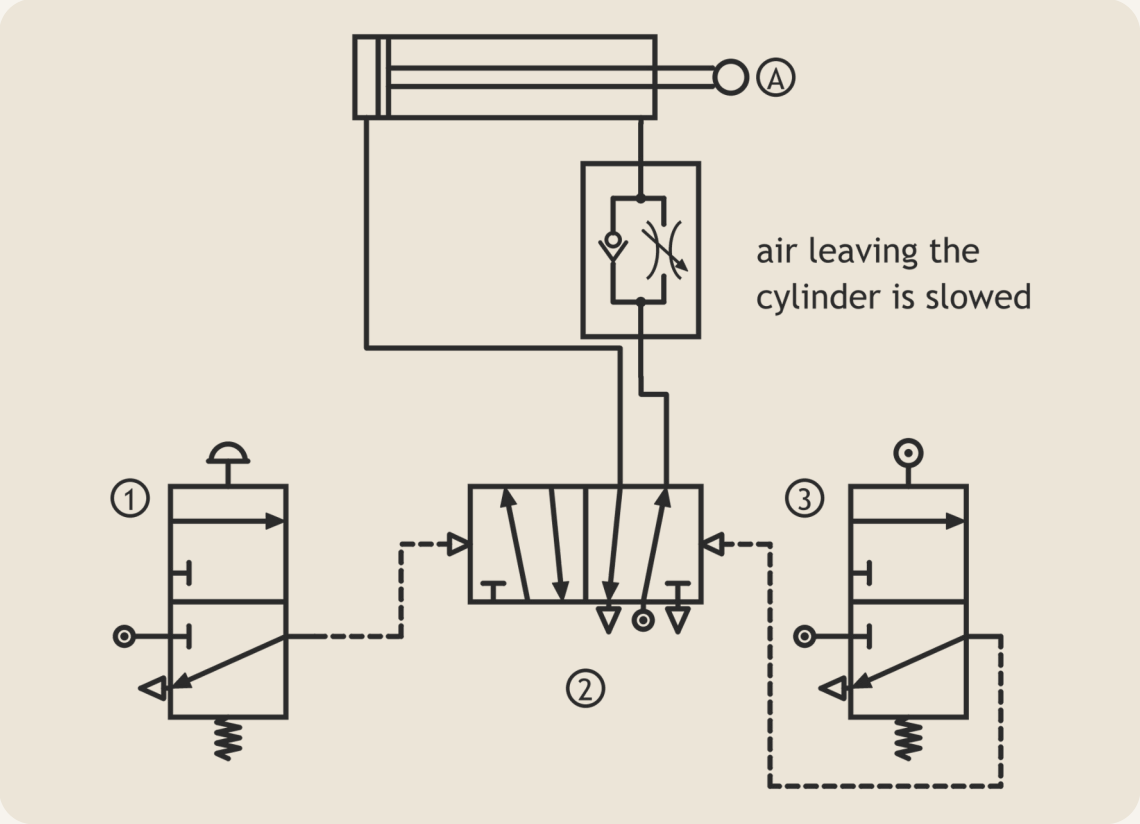
Changing the Delay (p.16)

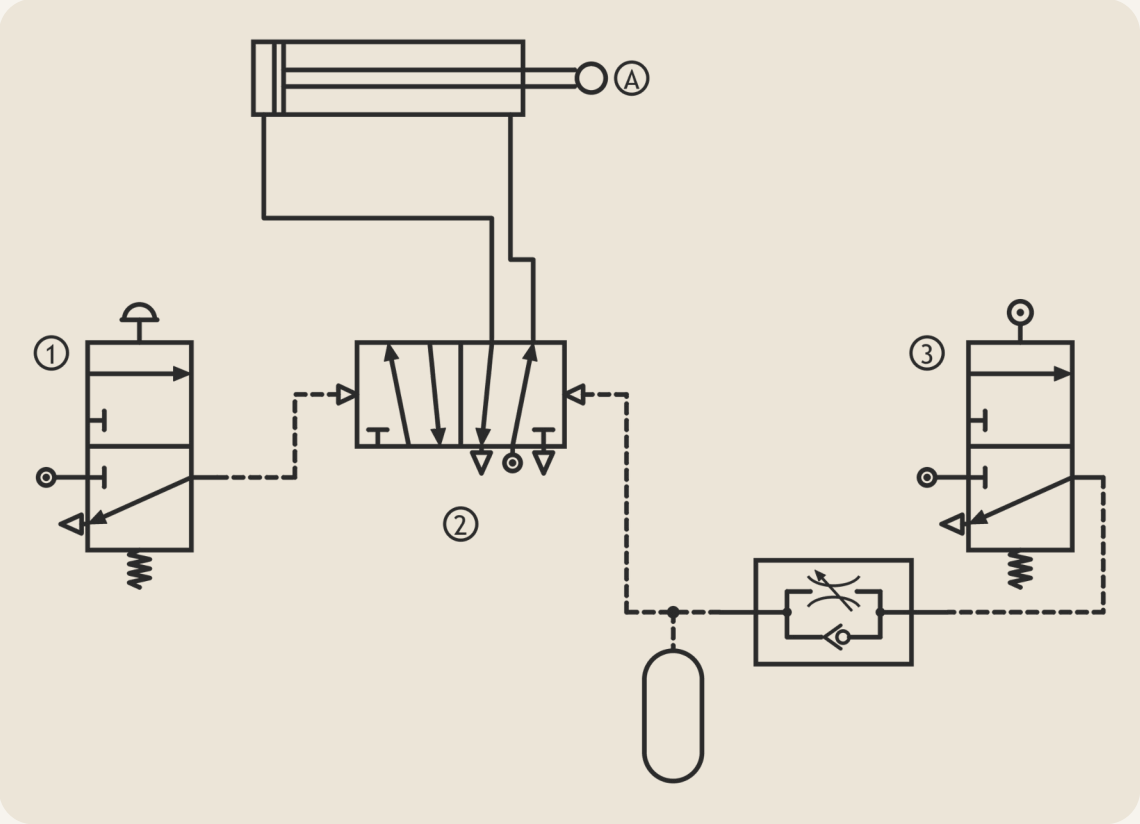
To make the delay **longer**: close the restrictor more, or use a **bigger** reservoir.

To make the delay **shorter**: open the restrictor more, or use a smaller reservoir.



Speed or Delay?





Speed (Pn4): it slows the **exhaust** air leaving the cylinder.

Delay (Pn5): it slows the air going **into** a reservoir.

Words That Get the Marks (p.16)

Draw

both symbols

Position

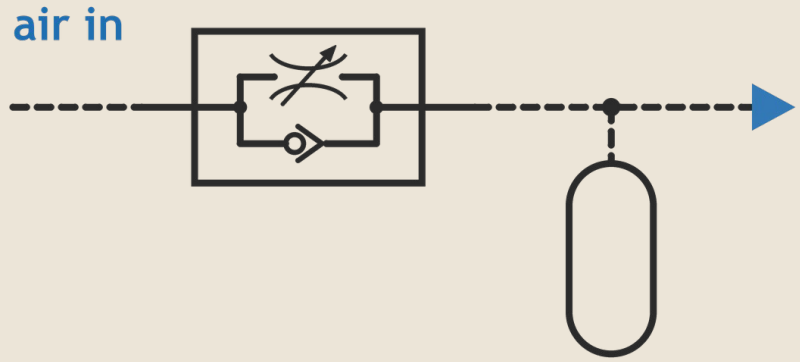
the restrictor **before** the reservoir

Right way round

it slows the air **into** the reservoir

Say

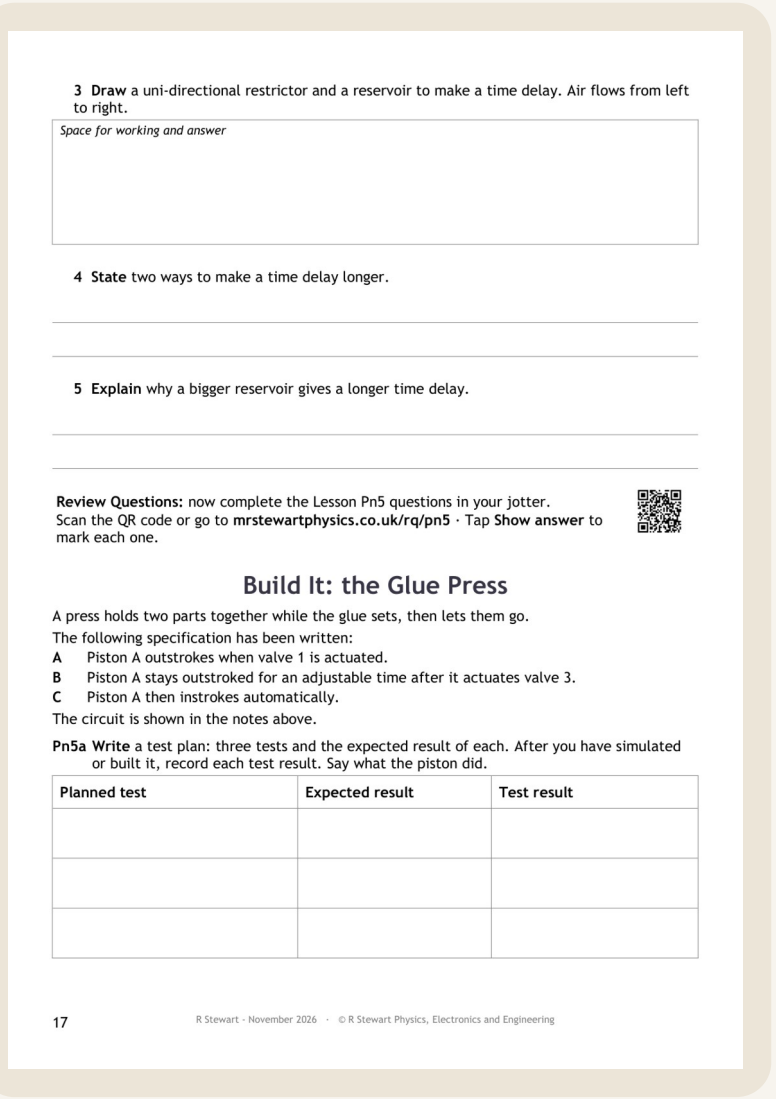
"a **time delay**"



Check Your Understanding (pp 16-17)

In your booklet

Answer Questions **1** to **5** on pages **16** and **17**.



3 **Draw** a uni-directional restrictor and a reservoir to make a time delay. Air flows from left to right.

Space for working and answer

4 **State** two ways to make a time delay longer.

5 **Explain** why a bigger reservoir gives a longer time delay.

Review Questions: now complete the Lesson Pn5 questions in your jotter. Scan the QR code or go to **mrstewartphysics.co.uk/rq/pn5** · Tap **Show answer** to mark each one.

Build It: the Glue Press

A press holds two parts together while the glue sets, then lets them go.

The following specification has been written:

A Piston A outstrokes when valve 1 is actuated.
B Piston A stays outstroked for an adjustable time after it actuates valve 3.
C Piston A then instrokes automatically.

The circuit is shown in the notes above.

Pn5a Write a test plan: three tests and the expected result of each. After you have simulated or built it, record each test result. Say what the piston did.

Planned test	Expected result	Test result

17 R Stewart · November 2026 · © R Stewart Physics, Electronics and Engineering

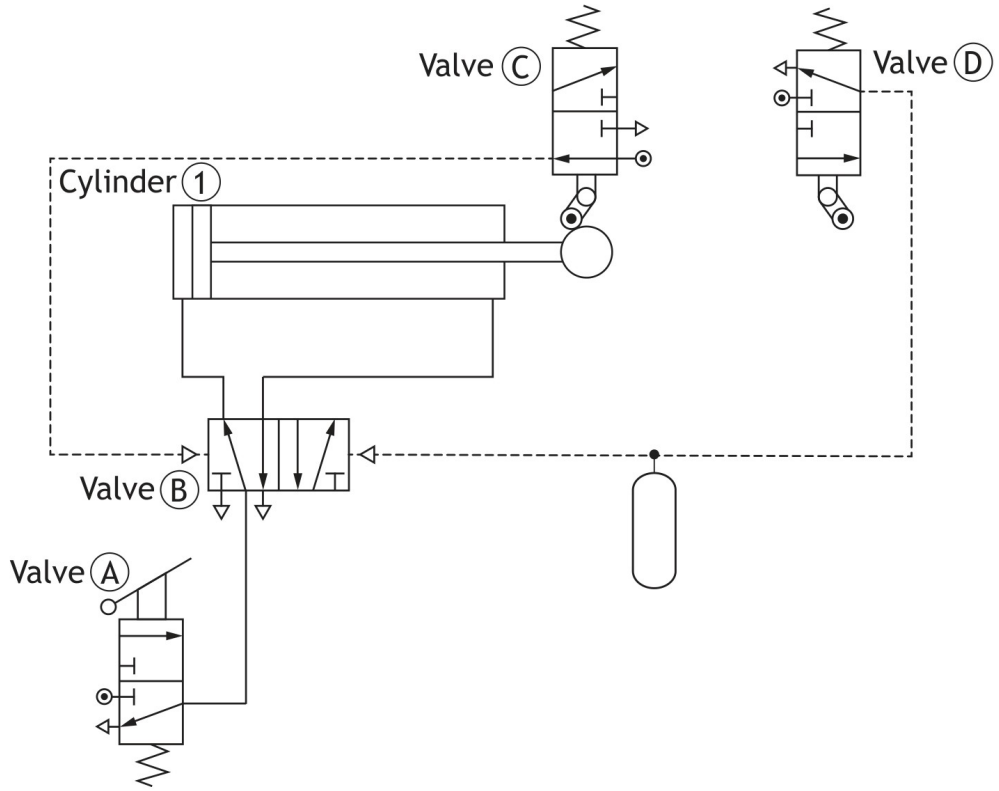
Answers: Check Your Understanding (pp 16-17)

Question	Answer
1	An **empty tank** that air must fill.
2	**Before** the reservoir, slowing the air going **into** it.
3	Restrictor slowing air to the **right**, then the **reservoir**.
4	**Close** the restrictor more. Use a **bigger** reservoir.
5	It takes **longer to fill**, **so** the valve changes **later**.

Exam: 2014 Section 2 Question 12 (a)

A pneumatic circuit used to punch holes in steel sheets is shown below.

(a) Draw a uni-directional restrictor in the correct position to form an **adjustable** time delay. **3**



Question 12 (a)

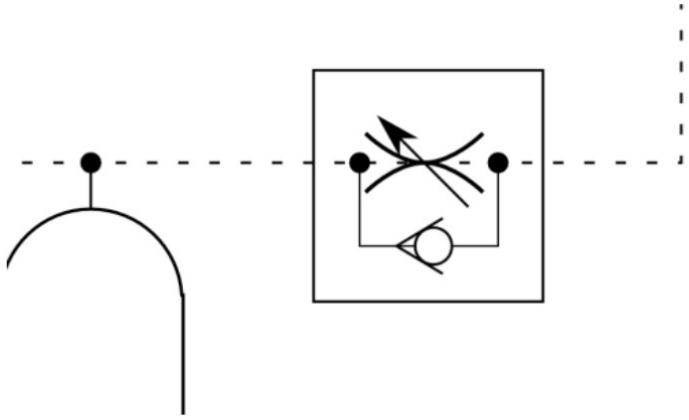
3 marks

Page 10 of the paper.

Sketch the circuit in your jotter.

Draw the restrictor in place.

Marking: 2014 Section 2 Question 12 (a)

Question			Expected Answer(s)	Max Mark	Additional Guidance
12	(a)			3	1 mark for correct symbol (ignore rectangle) 1 mark for correct position 1 mark for correct orientation

What Candidates Get Wrong

Three marks

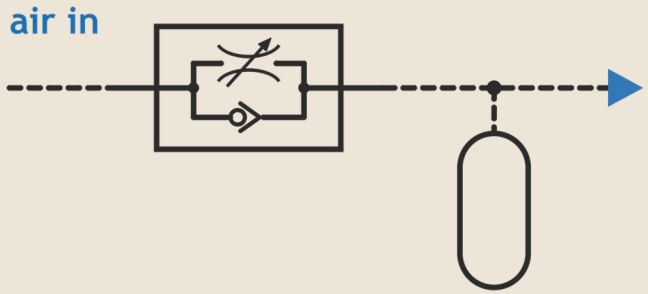
symbol, **position**, **orientation** (2014, 2017 MIs)

Order

the restrictor **before** the reservoir (2017 specimen MI)

Speed or delay?

many mixed them up (2023)



Review Questions (p.17)

Lesson Pn5 (online)

Answer the Lesson **Pn5** questions in your **jotter**.
Scan the QR or go to **mrstewartphysics.co.uk/rq/pn5**.
Tap **Show answer** to **mark** each one.

Scan me

Look back — by the end of this part… (p.16)



1. I can **describe** how a uni-directional restrictor and a reservoir make a time delay.
2. I can **draw** a uni-directional restrictor and reservoir the right way round.
3. I can **describe** how to make a time delay longer or shorter.
4. I can **describe** the operation of a circuit that has a time delay.

2026 N5 Engineering Science: scan the code and try these.

Q5

Page 6. Answer in your jotter.

Scan for the paper

Marking scheme

PART 2

Build It: the Glue Press

To plan, simulate, build and evaluate a pneumatic circuit, as the assignment asks.

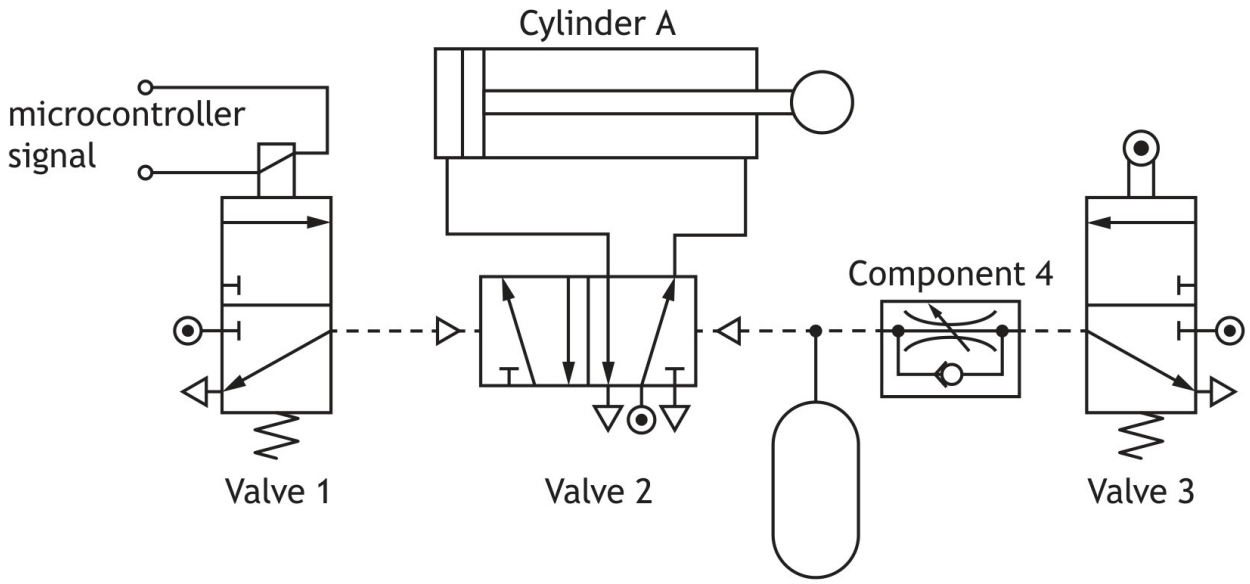
By the end of this part… (pp 17-18)



1. I can **write** a test plan with the expected result of each test.
2. I can **simulate** a circuit, then **build** it on the kit.
3. I can **evaluate** a circuit against its specification, using my tests.
4. I can **describe** a modification and **justify** it.

Exam: 2016 Section 2 Question 12 (a)

Part of the pneumatic circuit used in the model is shown below.



(a) Describe, using appropriate terminology, the operation of the circuit. **3**

When a signal is sent from the microcontroller _____

Question 12 (a)

3 marks

Page 14 of the paper.

Answer in your **jotter**: three marks, so **three** statements.

Marking: 2016 Section 2 Question 12 (a)

Question			Expected Answer(s)	Max Mark	Additional Guidance
12.	(a)		When a signal from the microcontroller actuates valve 1, pilot air actuating valve 2. This causes the piston to outstroke …air flow (through the uni-directional restrictor and reservoir) creating a time delay …. ..when valve 3 is actuated the piston will then instroke.	**3**	1 mark for each descriptive statement. 1 mark – outstroke condition 1 mark – time delay 1 mark – instroke condition

How It's Marked: One Mark per Statement

1 mark

outstroke: pilot air actuating valve 2 causes the piston to **outstroke**.

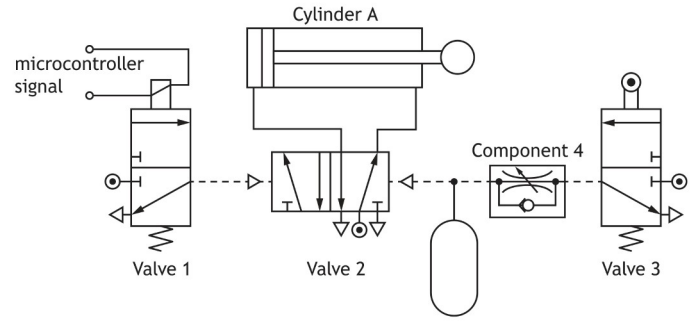
1 mark

time delay: air flow through the restrictor and reservoir.

1 mark

instroke: then the piston will **instroke**.

Part of the pneumatic circuit used in the model is shown below.



(a) Describe, using appropriate terminology, the operation of the circuit. **3**

When a signal is sent from the microcontroller _____

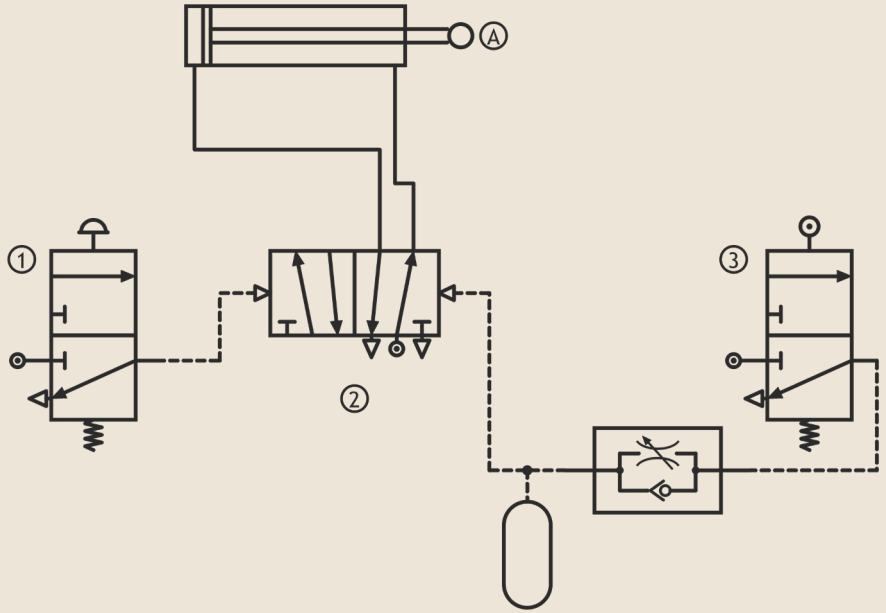
Build It: the Glue Press (p.17)

A press holds two parts while the glue sets.

A Piston A **outstrokes** when valve 1 is actuated.

B Piston A **stays outstroked** for an **adjustable** time after it actuates valve 3.

C Piston A then **instrokes** automatically.



Write a Test Plan (p.17)

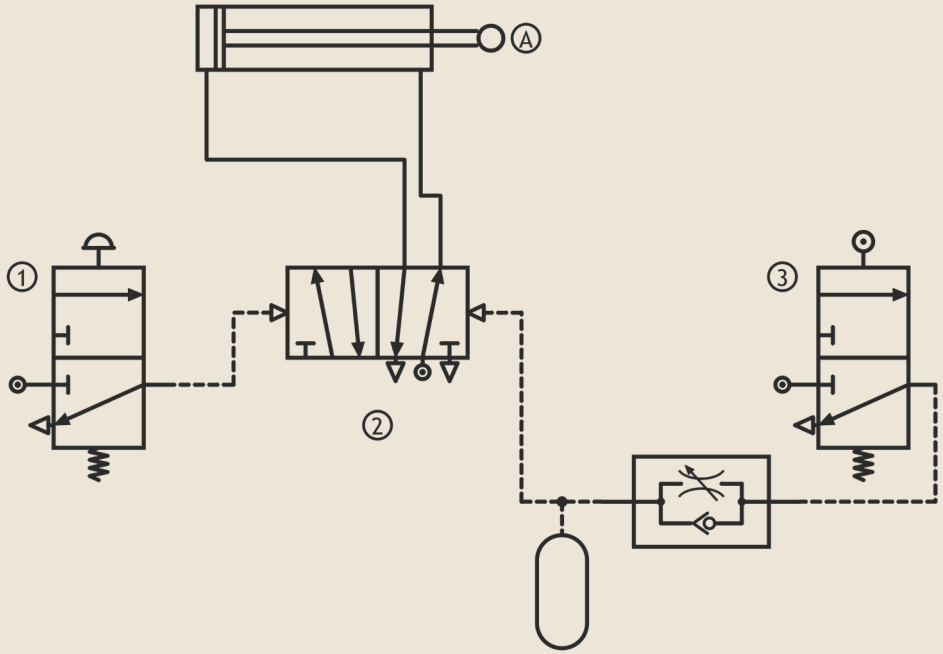


Planned test	Expected result	Test result
Actuate valve 1.	Piston A **outstrokes**.	
Time the delay. Adjust it. Time it again.	Piston A **stays out**. The time **changes**.	
Watch piston A after the delay.	Piston A **instrokes** automatically.	

Simulate It (p.18)

In your booklet

1. **Open** NoPressureSim. **Build** the circuit, as on page 8.
2. **Predict** what happens. **Test** it.
3. Air off: **build** it on the kit. Air on: **test**.



Scan me
NoPressureSim

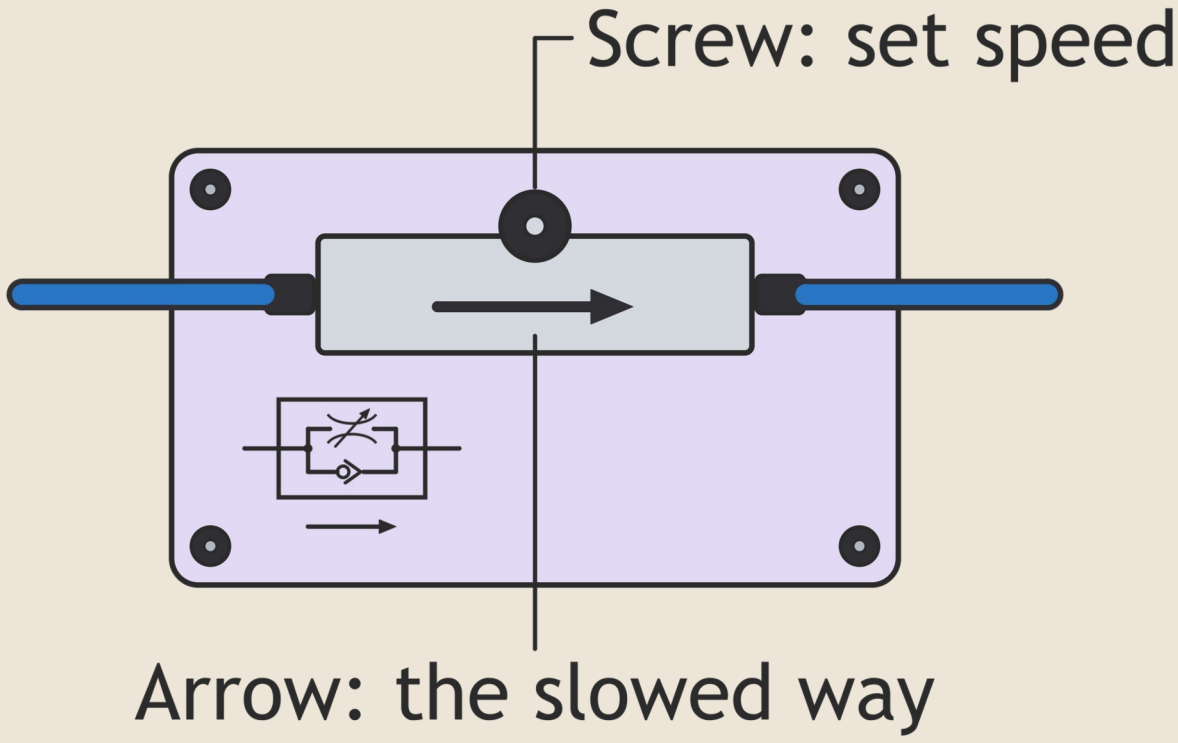
What to Collect

Collect one set per pair.

1. 3/2 push-button valve
2. 5/2 pilot/pilot valve
3. 3/2 roller valve
4. Double-acting cylinder
5. Restrictor, reservoir and

T

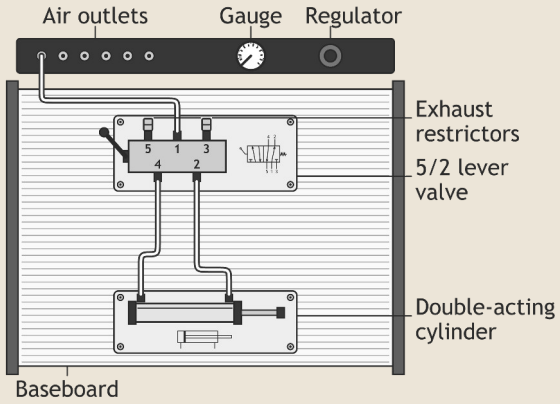
6. Ten blue tubes
7. Goggles each



Build It on the Kit (p.18)

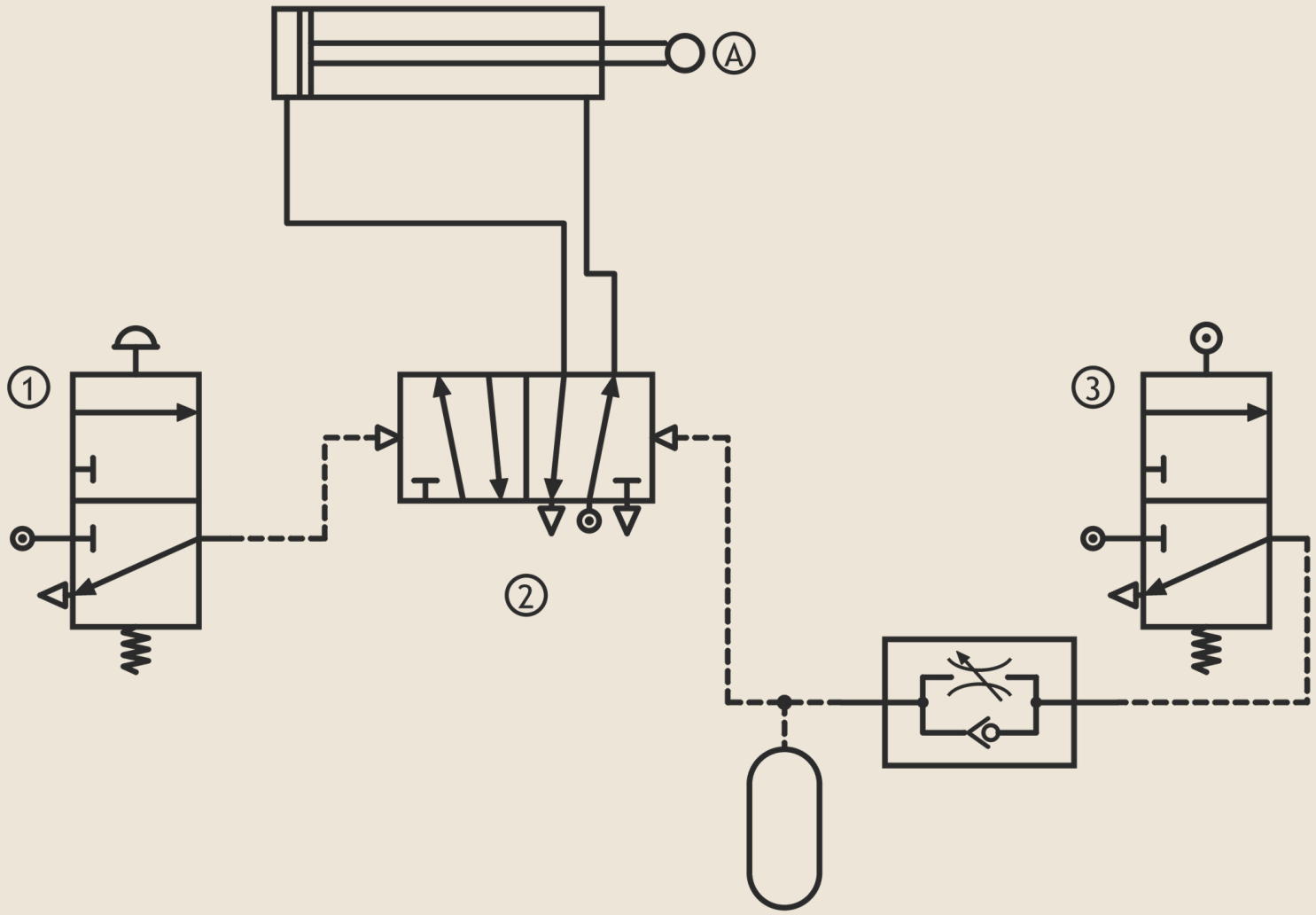
On the kit

Build it, air **off**.
Five reservoirs: the first **five** pairs signed off build first.
Air on. **Test** and **record**.



SAFETY 0 bar to build. 1 to 2 bar to test.

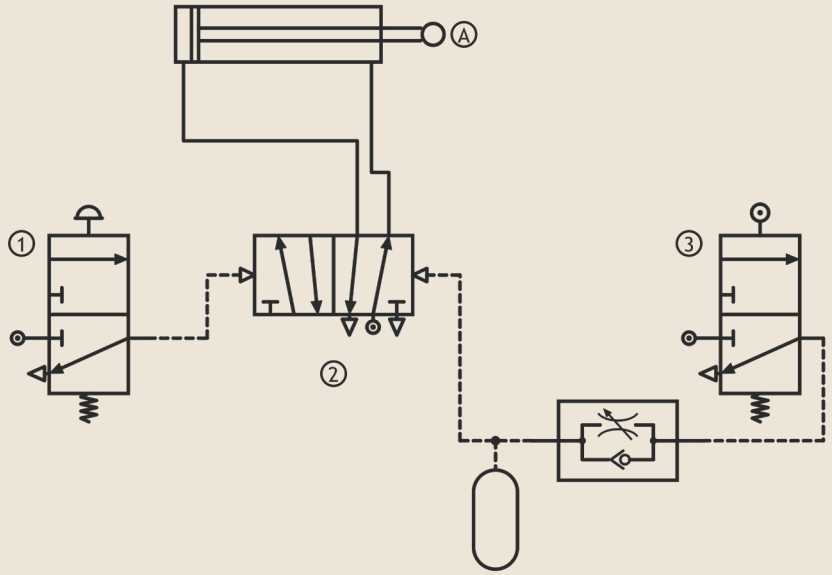
Build It: Bench Card 1



Evaluate (p.18)

In your booklet

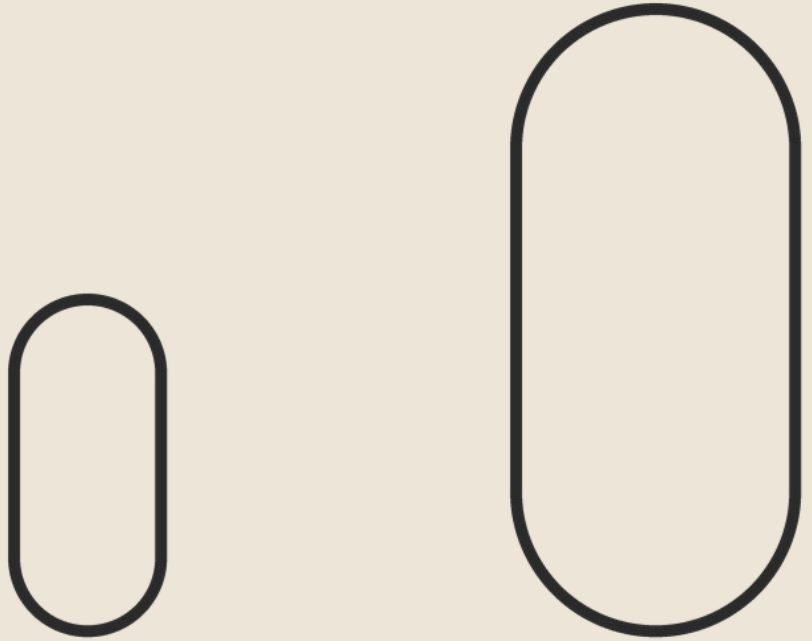
For each specification point, **state** if it was met.
Explain why, using your **test results**.
Say what you did **and** what the piston did.



Improve and Justify (p.18)

In your booklet

Some glues need a **much longer** time to set.
Describe a modification that would solve this.
Justify it: say **why** it solves the problem.



Answers: Evaluate (p.18)

Point	Met?	Explanation
A	Yes	**In test 1**, valve 1 was **actuated**, pilot air **changed the state** of valve 2, **so** piston A **outstroked**.
B	Yes	**In test 2**, the reservoir **filled slowly**, **so** piston A **stayed out**. The restrictor **adjusted** the time.
C	Yes	**In test 3**, the pressure in the reservoir **changed the state** of valve 2, **so** piston A **instroked**.

Answers: Improve and Justify (p.18)

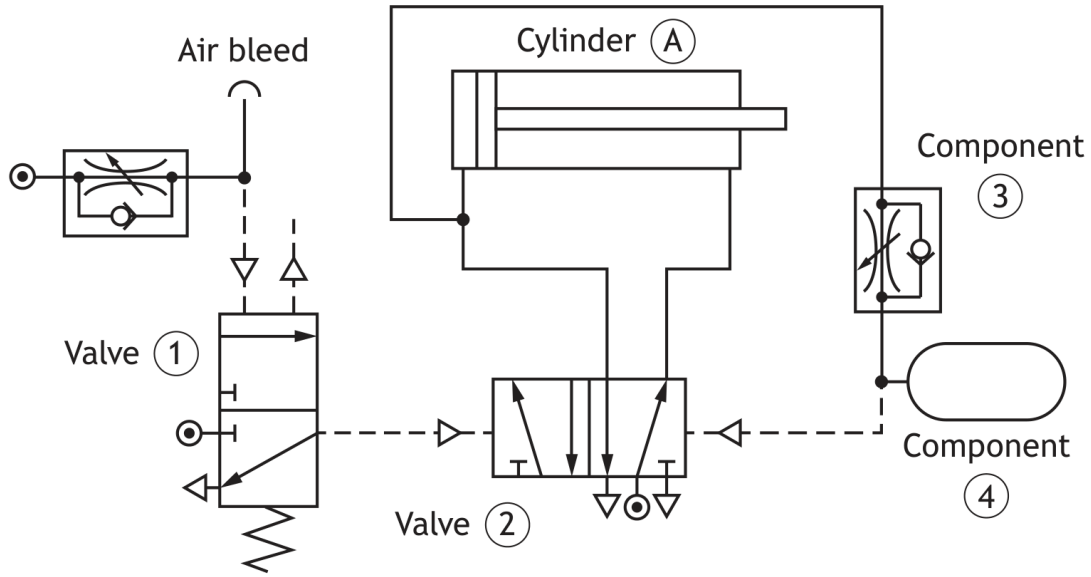


	Answer
Modification	Use a **bigger** reservoir (or **close** the restrictor more).
Justification	A bigger reservoir takes **longer to fill**, **so** the time delay is **longer**.

If Time: 2018 Section 2 Question 15 (c)

Portions of the pastry travel along a conveyor belt where a pneumatic piston presses them into pie casings.

The pneumatic circuit shown below operates the piston when the pastry is sensed in position.



(c) Describe, using appropriate terminology, the operation of the pneumatic circuit, shown above. **3**

When the air bleed is covered valve 1 is actuated.

Question 15 (c)

3 marks

Page 28 of the paper.

Components **3** and **4**: a **uni-directional restrictor** and a **reservoir**.

Marking: 2018 Section 2 Question 15 (c)

Question			Expected response	Max mark	Additional guidance
15.	(c)		*When the air bleed is covered valve 1 is actuated...* ...pilot air is sent to Valve 2 causing the piston in Cylinder A to outstroke… …air is also sent to (Components 3 and 4) creating a time delay... ...pilot air is sent to Valve 2 again causing the piston in Cylinder A to instroke.	**3**	Must be descriptive responses using appropriate terminology 1 mark for Valve 2 causing piston to outstroke 1 mark for time delay before instroking (second piston movement) 1 mark for Valve 2 causing piston to instroke

What Candidates Get Wrong

Role and effect

say what the restrictor **does** (2023)

Speed or delay?

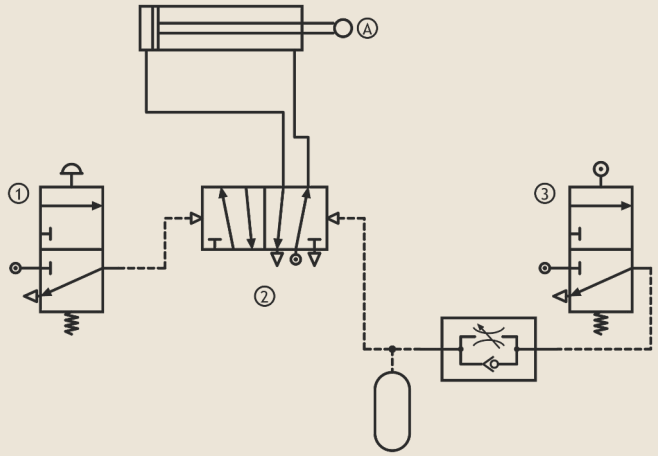
at the cylinder it is **speed** (2023)

One mark each

outstroke, **delay**, instroke (2016, 2018)

Terminology

sentences, not bullets (2023)



Look back — by the end of this part… (pp 17-18)



1. I can **write** a test plan with the expected result of each test.
2. I can **simulate** a circuit, then **build** it on the kit.
3. I can **evaluate** a circuit against its specification, using my tests.
4. I can **describe** a modification and **justify** it.

Scan: page 6

Pneumatics

- symbols and operation of standard pneumatic components (restrictor, uni-directional restrictor, reservoir, 5/2 valve and actuators: diaphragm and solenoid)
- pneumatic time delay circuits
- calculating relationships between force, pressure and area in single-acting and double-acting cylinders
- controlling speed and force

Credits (1 of 3)

Drawings: Pneumatic symbols and circuits drawn by R Stewart in the Qualifications Scotland paper style; CP2080 kit drawings by R Stewart.

Starter: 2019 N5 Engineering Science, © Qualifications Scotland (SQA). 2019 question paper and marking instructions; 2025 N5 Engineering Science, © Qualifications Scotland (SQA). 2025 question paper and marking instructions; 2026 N5 Engineering Science, © Qualifications Scotland (SQA). 2026 question paper and marking instructions

Exam: 2014 Section 2 Question 12 (a): © Qualifications Scotland (SQA). 2014 National 5 Engineering Science Section 2, Q12

Marking: 2014 Section 2 Question 12 (a): © Qualifications Scotland (SQA). 2014 National 5 Engineering Science Section 2, Q12 marking instructions

Credits (2 of 3)

What Candidates Get Wrong: © Qualifications Scotland (SQA). National 5 Engineering Science course report 2023; 2014 S2 Q12(a), 2017 S2 Q11(b) and 2017 specimen S2 Q16(a) marking instructions; © Qualifications Scotland (SQA). National 5 Engineering Science course report 2023; 2016 S2 Q12(a) and 2018 S2 Q15(c) marking instructions

Screenshots and page figures: Booklet 6, Pneumatics (N5 Engineering Science, R Stewart, November 2026): page images.

Exam: 2016 Section 2 Question 12 (a): © Qualifications Scotland (SQA). 2016 National 5 Engineering Science Section 2, Q12

Marking: 2016 Section 2 Question 12 (a): © Qualifications Scotland (SQA). 2016 National 5 Engineering Science Section 2, Q12 marking instructions

How It's Marked: © Qualifications Scotland (SQA). 2016 National 5 Engineering Science Section 2, Q12 marking instructions

Credits (3 of 3)

If Time: 2018 Section 2 Question 15 (c): © Qualifications Scotland (SQA). 2018 National 5 Engineering Science Section 2, Q15

Marking: 2018 Section 2 Question 15 (c): © Qualifications Scotland (SQA). 2018 National 5 Engineering Science Section 2, Q15 marking instructions

Course specification: © Qualifications Scotland (SQA). N5 Engineering Science course specification v2.0, p.6

R Stewart - November 2026